PEA
PROVINCIAL ELECTRICITY AUTHORITY

Koh Tao Microgrid — Dispatch Order

17:30 Thu 27-Mar-2026 | Solve `#0503-1728` | Valid until next re-solve **18:00**

NOW: Load **12.8 MW** · Grid **4.0 MW (capped)** · BESS **61% SoC, discharging 7.2 MW** · Diesel **OFFLINE**

🔴 NEXT ACTION — Start Diesel Genset

Execute by	**17:45** (T-15 from now)
Command	START → synchronize → close breaker → ramp to **3.0 MW by 17:55** → **6.5 MW by 19:00**
Reason	BESS hits 10% floor at ~19:45 without DG; load peaks 14.0 MW at 19:00
If missed	Declare emergency. Begin manual load-shed per **SOP-MG-04**

Pre-Start Checklist (complete before 17:43)

- ☐ Day tank fuel level ≥ 80% · transfer pump in AUTO
- ☐ Lube-oil temp ≥ 40 °C · jacket water ≥ 50 °C (pre-heat ON)
- ☐ AVR in AUTO · Governor in DROOP (4%) · PMS in AUTO-PARALLEL
- ☐ Sync key inserted · check-sync relay armed
- ☐ DG breaker rack-in confirmed · protection (51V, 32, 40, 27/59) green

Sync / Parallel (at 17:45 + sync window)

Close DG breaker only when **all three** met on synchroscope: $\Delta V \leq \pm 5\%$ · $\Delta f \leq \pm 0.2$ Hz · $\Delta\varphi \leq \pm 10°$ (slip slow, 12 o'clock)

Ramp Rate

Genset **+1.0 MW/min max**. Do not step-load. Hold 3.0 MW for 60 s after sync before further ramp.

Hourly Setpoints (17:00 – 22:00)

Window	Load	Grid	BESS	Diesel	SoC EoH
17:00–17:59	12.8	4.0	DIS 7.2	OFF	55%

Window	Load	Grid	BESS	Diesel	SoC EoH
18:00–18:59	**13.5**	**4.0**	**DIS 5.5**	**3.0 MW** *(just synced)*	**52%**
19:00–19:59	**14.0**	**4.0**	**DIS 3.5**	**6.5 MW**	**46%**
20:00–20:59	13.8	4.0	DIS 3.8	6.0 MW	40%
21:00–21:59	12.5	4.0	DIS 2.5	6.0 MW	37%

All values in MW. Re-solve at 21:00 will reassess DG de-load.

Hard Limits — Do Not Breach

Limit	Value	If breached
BESS SoC floor	**10%** (warn at 15%)	Trip BESS, raise DG to max, request emergency re-solve
Spinning reserve	**≥ 15% of load**	Re-solve immediately; do not accept new dispatch
Grid intake (33 kV cable)	**4.0 MW hard cap**	Cable protection trips at 4.4 MW — back off setpoint
DG min stable load	**3.0 MW**	Below this → load on resistive bank or take DG offline

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Ack diesel-start authorisation by 17:43 | Supervisor ext. **201**